Thursday, March 27, 2025

LEC33 - Power Series Continued

Warm Up Problem

Does the series $\sum_{n=1}^{\infty} a\left(\frac{1}{2}\right)^{n-1}$ converge, and if so, what is its sum?

$\Rightarrow$ Geometric series where $r = \frac{1}{2} < 1$

$\Rightarrow$ Converges to $\frac{a}{1-1/2}$

Example

Find a power series representation of each function, and find its radius of convergence, R

i) $f(x) = \frac{1}{1+x}$

$\Rightarrow \frac{1}{1-(-x)}$, which is the result of a geometric series

$\Rightarrow \sum_{n=0}^{\infty} (-x)^n$, where $|-x| < 1 \Rightarrow -1 < x < 1$

ii) $g(x) = \frac{1}{1+x^2}$

$\Rightarrow \frac{1}{1-(-x^2)}$, which is the result of a geometric series

$\Rightarrow \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n}$, where $|-x^2| < 1 \Rightarrow -1 < x < 1$

iii) $h(x) = \arctan x$

$\Rightarrow \frac{d}{dx}[\arctan x] = \frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$

$\Rightarrow \arctan x = \int \sum_{n=0}^{\infty} (-1)^n x^{2n} dx = \sum_{n=0}^{\infty} (-1)^n \int x^{2n} dx$

$\Rightarrow \sum_{n=0}^{\infty} (-1)^n \cdot \frac{x^{2n+1}}{2n+1}$, where $|x| < 1 \Rightarrow -1 < x < 1$

Example

a) What are the sum and radius of convergence of $f(x) = \sum_{n=0}^{\infty} \frac{x^n}{4^n}$?

$\Rightarrow$ Geometric series where $r = \frac{x}{4}$

$\Rightarrow$ Converges to $\frac{1}{1-\frac{x}{4}}$, where $\left|\frac{x}{4}\right| < 1 \Rightarrow -4 < x < 4$

$\Rightarrow$ Therefore, the sum is $\frac{1}{1-\frac{x}{4}}$, and $R = 4$

b) $\sum_{n=0}^{\infty} \frac{nx^{n-1}}{4^n}$ is the derivative of the series from (a). What is its sum and radius of convergence

$\Rightarrow$ We know that $\sum_{n=0}^{\infty} \frac{x^n}{4^n} = \frac{1}{1-\frac{x}{4}}$, therefore $\frac{d}{dx}\left[\sum_{n=0}^{\infty} \frac{x^n}{4^n}\right] = \frac{d}{dx}\left[\frac{1}{1-\frac{x}{4}}\right]$

$\Rightarrow \sum_{n=0}^{\infty} \frac{nx^{n-1}}{4^n} = \frac{d}{dx}\left[\frac{1}{1-\frac{x}{4}}\right] = \frac{1}{4(1-\frac{x}{4})^2}$, where R remains unchanged

Properties of Power Series

If given two congruent power series $\sum_{n=0}^{\infty} c_n x^n$ and $\sum_{n=0}^{\infty} d_n x^n$ it is possible to create a new series via the following operations:

1. Addition term by term: $\sum_{n=0}^{\infty} (c_n + d_n) x^n$
2. Subtraction term by term: $\sum_{n=0}^{\infty} (c_n - d_n) x^n$
3. Multiply by bx^m: $bx^m \sum_{n=0}^{\infty} c_n x^n = \sum_{n=0}^{\infty} bc_n x^{n+m}$
4. Compose with bx^m: $\sum_{n=0}^{\infty} c_n (bx^m)^n = \sum_{n=0}^{\infty} c_n b^n x^{mn}$
5. Multiply the two series term by term (like infinite polynomials)
6. Differentiate term-by-term
7. Integrate term-by-term

Note: In all but 4, the radius of convergence does not change

Example

We know from before that $\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$, with $R = 1$. Use this to find a series that sums to $\frac{\pi}{6}$

$\Rightarrow \arctan x = \frac{\pi}{6} \Rightarrow x = \frac{\sqrt{3}}{3}$ $(\tan \frac{\pi}{6})$

$\Rightarrow \arctan\left(\frac{\sqrt{3}}{3}\right) = \frac{\pi}{6} \Rightarrow \sum_{n=0}^{\infty} (-1)^n \frac{\left(\frac{\sqrt{3}}{3}\right)^{2n+1}}{2n+1}$

$\Rightarrow$ Therefore, $\frac{\pi}{6} = \sum_{n=0}^{\infty} (-1)^n \frac{\left(\frac{\sqrt{3}}{3}\right)^{2n+1}}{2n+1}$